Fatemeh Fallahnejad

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EDUCATION

University of California, San Diego — *Sept 2026 – June 2028 (expected)*
B.S. in Computer Science - Chancellor's Associates Scholarship (CASP) Scholar — *La Jolla, CA*

San Diego Miramar College — *Graduated June 2026*
A.S. Computer Science · A.S. Chemistry Studies · A.A. Mathematics Studies · A.A. Social & Behavioral Sciences — *GPA: 4.00*
Research Excellence Award, Honors Scholar, Dean's List

Relevant coursework:
- Data Structures in C++ (CSE 12)
- Computer Organization & Assembly Language (CSE 30)
- Discrete Mathematics (CSE 20)
- Foundations of Data Science (DSC 10)
- Structure & Interpretation of Computer Programs (CSE 11)
- Calculus I–III · Linear Algebra & Differential Equations

EXPERIENCE

Data Science & Computational Biology Intern — *June - Aug 2026*
Vividion Therapeutics — *San Diego, CA*
- Developed **software components and the user interface** for an internal RNA-sequencing analytics platform, and deployed it to a server for organization-wide use by research scientists.
- Acquired the required domain knowledge to translate biology analysis workflows and built **Python backend services** that process sequencing data and serve results to the web interface, without additional need for coding.
- Contributed to training **language models** for **protein representation** learning with the cheminformatics team.

Undergraduate Researcher - NSF REU in Interdisciplinary AI — *Sept 2025 - May 2026*
Cottrell Lab, Dept. of Computer Science & Engineering, UC San Diego — *La Jolla, CA*
- Benchmarked and built the **evaluation pipeline** and **held-out test sets** used to **validate model performance** across **seven generative architectures** (VAEs, GANs, transformers, and LSTMs) in **Python and PyTorch**. Data representation drove performance more than architecture: 100% chemical validity for one model class versus 81.6%.
- Ran **systematic ablations** isolating one variable at a time; expanding a model's latent space (32 to 256) dimensions lifted exact-match reconstruction from 0.440 to 0.650, a 48% relative gain.
- Architected an end-to-end **graph diffusion pipeline** (spectral transformer ⟶ graph tokenizer ⟶ discrete diffusion model) generating candidate molecular structures from NMR spectra.
- **Testing protocol** on a 150-molecule held-out test set: 65.3% of outputs were chemically valid, and 10% differed from every retrieved candidate. **Stack: RDKit, MOSES, molecular reconstruction, and Tanimoto metrics.**

Research Scholar - Computational Physics & Astronomy — *June - Sept 2025; Aug 2026*
Pathway to STEM Success (PASS) Summer Scholars Program — *Mount Wilson Observatory, CA*
- Applied **large language models (LLM APIs)** to automate telescope imaging and observation sequencing through N.I.N.A., reducing manual operator intervention during observing runs. Conducted independent research on binary star systems using remote observations; authored a technical paper for the *Journal of Double Star Observations*.

Student Success Specialist — *Aug 2025 - June 2026*
San Diego Community College District - MESA Program — *San Diego, CA*
- Tutored undergraduates in mathematics, engineering, and computer science; coordinated events with faculty, alumni, and external partners.

Extended Day Program Aide & Tutor — *April 2024 - June 2025*
Cajon Valley School District — *El Cajon, CA*
- Led and supervised 30-40 K–12 students; tutored mathematics, coding, biology, chemistry, and physics.

SKILLS

Languages & Tools: Python, C++, Java, R, Bash, Linux, Jupyter · **AI/ML:** LLM APIs, PyTorch, TensorFlow, neural networks, model training
Software & Deployment: Backend development, web user interfaces, REST APIs, server deployment, data pipelines.
Communication: Technical presentations, cross-disciplinary collaboration, Socratic tutoring; bilingual in English and Persian